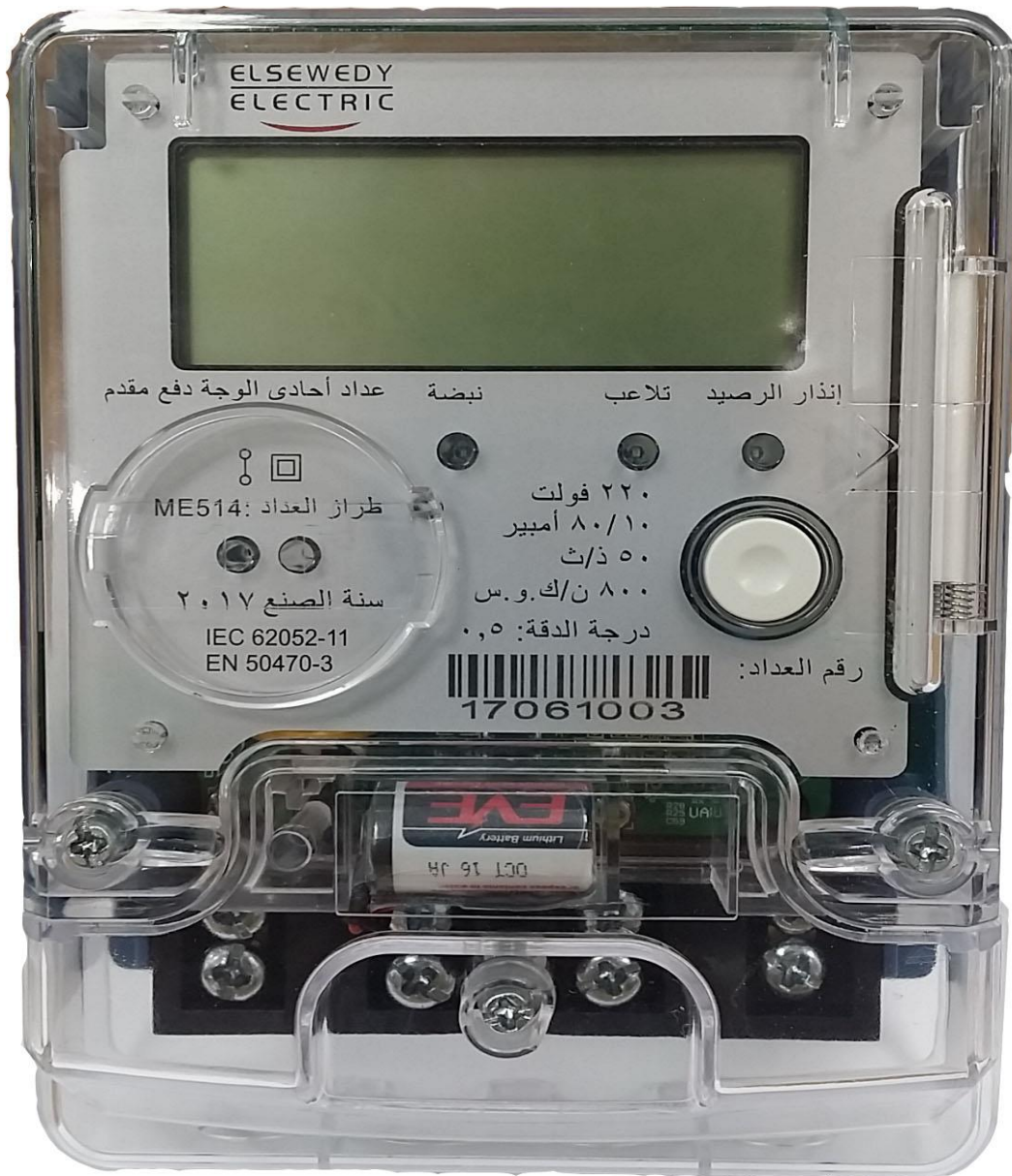


ME514

Single Phase Prepaid Meter with CPU Card



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1 Overview

ME514 is a type of single phase electronic multi-function energy meter, which adopts advanced micro electronic technology and SMT production process. The operation manual is mainly used for development and design, production instruction, quality assessment, as well as technical support, on the requirements in IEC62052-11 and EN50470-3

The meter can measure single phase energy of specified voltage and frequency; it is also equipped with multiple functions which are step tariff, Billing, load limit alarm, the remaining days alarm and anti-tamper function, etc. The meter circuit is consists of power supply, MCU, metering, CPU card circuit, clock, LCD display, storage, relay drive circuit, battery circuit, etc.

2 Functions and Feathers

- ☆ The CPU card prepayment mode means pay first before energy use.
- ☆ Metering accumulative active energy, and the minimum metering energy is 0.01kWh.
- ☆ Metering the relay failure electricity, and the relay failure consumption is included in the total consumption
- ☆ Metering reverse energy, and the reverse energy is included in the total energy consumption
- ☆ Display the instantaneous current, voltage, power, power factor
- ☆ LCD display with Arabic language
- ☆ LED alarm
- ☆ With remaining energy alarm threshold and trip-out threshold, to remind users to recharge
- ☆ With load control function
- ☆ Support double-loop metering to avoid tampering.
- ☆ Meter can detect cover open and terminal cover open even meter power off.
- ☆ Support optical communication so as to read and set meter parameters.

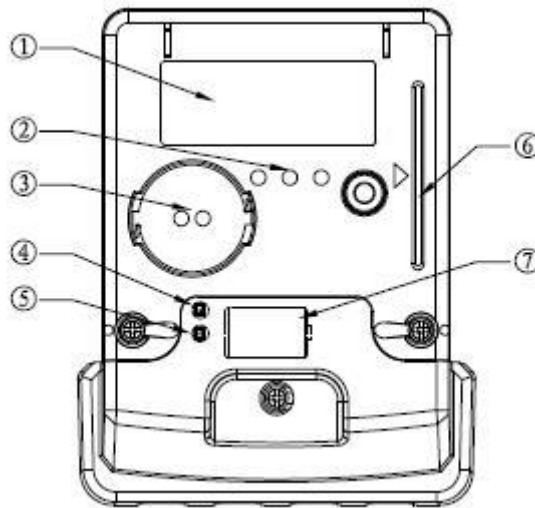
3 Technical specifications

Basic voltage	220V
Basic current	10(80)A
Frequency	50 Hz (±2%)
Accuracy class	Class C
Burden	Active power: 0.7W

Pulse constant	800imp/kWh
Outline dimension	L226.8mm×W131mm×H72.5mm
Temperature	-5 °C 70 °C
Temperature Limit	-5 °C 85 °C
Humidity	Up to 95 %

※Note: The actual information can refer to the nameplate on meter.

4 Meter profile

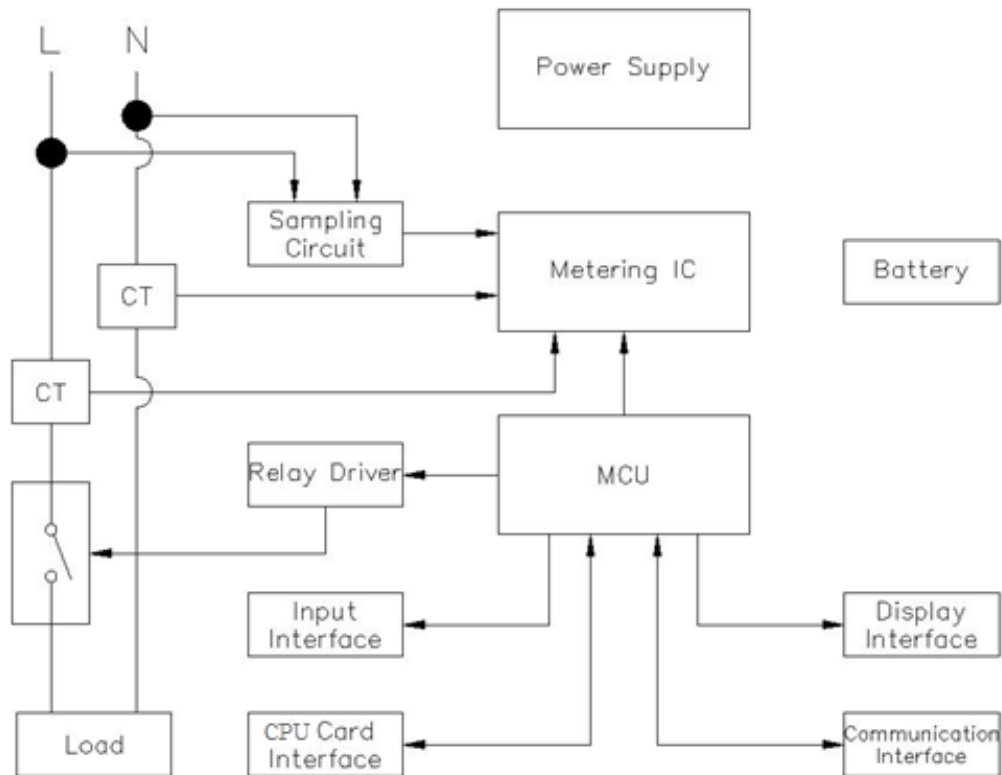


【Figure 4.1】 Note: There may be slight differences, please refer to the physical samples

- 1) LCD screen: Used for meter information display
- 2) LED indicator: Pulse, Alarm, and Low Credit. (From left to right)
- 3) Optical communication port
- 4) Meter cover detect switch
- 5) Terminal cover detect switch
- 6) CPU card operation interface
- 7) Battery

5 Operation principles

The microprocessor unit (MCU) for calculate the voltage and current signal which is sampling from CT, after calculation with input signals then achieves the energy metering and real-time instantaneous value reading. The LED and LCD is display the meter status and data information. The meter also has high accuracy clock circuit (optional) and backup battery for power supply. Meter data can be read and written through the communication port.



【Figure 5.1】

6 Functions

6.1 Metering function

Meter can measure the total forward and reverse active energy; reverse energy is record into the forward active energy. It can record the historical data of last 12 monthly billing periods at most and displayed on the LCD screen. Data is saved in the non-volatile memory and can be stored for more than 15 years without power supply.

6.2 Billing function

Each month has only one billing time, and 12 months' historic billing data can be recorded at most, these data are the import and export active energy of each tariff. Billing time can be set at the time of any date between 1st and 28th. When the billing time, meter will auto-billing and save the data (the energy consumption and the occurred billing time) into last month. Even if the meter is at the billing time and without power (not over 12 months), it will auto-billing over these days or months.

If the power consumption of the current month is 0 when billing, it will be deducted the special tax fee.

6.3 Tariff function

6.3.1 Time of using

Meter has the reserved tariff information and activation time, the activation time could be configurable. Meter is support the century calendar, RTC and auto-switching to leap year. If the tariff index number of any time interval is over than the tariff number, tariff will be default to tariff 1. Meter is always worked on the single tariff mode when at any 1 tariff of any 1 time interval of any 1 time zone.

6.3.2 Step tariff

The maximum is 9 steps and 10 tariffs. Meter has the reserved step schedule and activation time, the activation time could be configurable. Each step has a related step tariff. When deduct fee unit is 1kWh, the remaining credit will deduct once with 1kWh energy consumption. Step and tariff value can be configurable by CPU card; Meter is work on single-tariff when there is no step and only one step tariff.

TOU and step tariff could be switch to each other. The switch time could be configurable by the CPU card. When tariff has switched from TOU to step tariff, the energy consumption which is since the billing date will be calculated into step tariff. For example, when the energy consumption is 150KWh which is since the billing date, if TOU is switches to step tariff (100 to 200 is the second step), then the credit will be deducted according to the second tariff. The meter will record it as an event with time stamp and related total energy consumption information.

6.4 Measurement function

Meter can measure voltage, current, active power, power factor and other instantaneous parameters.

- 1) Active power : Data refresh by each second. Measuring range is $5\%P_b \sim P_{max}$ (P_b is the rated active power , P_{max} is the max active power) . Power unit is kW, measuring error (accuracy error) should not exceed $\pm 0.5\%$.
- 2) Voltage : Data refresh by each second. Measuring range is $80\% \sim 115\% U_n$.
- 3) Current : Data refresh by each second. Measuring range is $5\%I_b \sim I_{max}$.

6.5 Communication function

Optical communication

Support optical comm. for local data reading and writing, comply with IEC61107 protocol.

CPU card function

When insert the specified CPU card into meter, total energy consumption, energy consumption of each tariff, remaining credit, purchased times, meter state and other information will be automatically written back to CPU card, so that the vending management system can check the meter status.

If necessary, other data that needs to be return back also could be specified by vending system when make card.

6.6 Event record function

Meter is support the event record function, the event is such as meter cover open, terminal cover open, battery under voltage, relay error etc. Each kind of event can up to 10 times record with time stamp and meter status, this function is for preventing tampering.

Note: Factory resetting card can clear event logs.

History energy consumption record

Auto-record the energy consumption (accumulative energy consumption of current month) of last 12 months, and generate 12 monthly records

If miss record the related data when power off at billing time, meter will automatically fill it and default it as to 0.

Power failure record

Record the total power failure times since opening account, and generate logs with time stamp of last 5 times.

Time adjustment record

Record the total time adjustment times, and generate logs with time stamp of last 5 times.

Purchasing record

Record the total purchasing times, and generate logs with time stamp of last 5 times.

Tariff switch record (TOU and step tariff)

Record the total switch times, and generate logs of last 5 times. The record information has total energy consumption, remaining credit, energy consumption of current month, energy consumption of tariff 1, tariff 2, tariff 3 and tariff 4, as well as the switch time stamp.

Battery under voltage record

Record the total battery under voltage times, and generate the logs with time stamp of last 10 times.

Terminal cover and top cover open record

Record the total terminal cover open times, and generate the logs with time stamp of last 5 times.

Record the total top covers open times, and generates the logs with time stamp of last 5 times.

Current unbalance record

When the current difference value between the fire and neutral line is exceed the threshold, it will generate the logs of last 5 times with the start time and end time at the same time.

Note:

1. When power on, the current status is switch from unbalance to balance, record the time as the end time of this unbalance;
2. If the current unbalance is not released when power off, recording the power off time as the end time.

Current reverse record

Record the starting time of Current reverse and generate related records of last 5 times.

Fault remove record

Record the time and operator number of fault of last 10 times.

Relay error record

Record the starting time of relay error of last 5 times.

Inset card record

When the meter accept card, meter can record the card type, card number, inserting date and time. This record is maximum 10 times.

6.7 Over load

In case of over load last for 5 minutes, the relay will open, then after 5 mins, the relay will be automatically close. If relay is the 3th trip off as the over current, the relay will not close again. It's only use the specified card to remove this alarm event

Note: If the over load times is less than 3 times in this month, the count will be automatically reset to 0 when the next month is start (at the monthly billing time).

Each day could be programmed to 48 time intervals with different current limit value.

Meter judge over load by current or power can be configured by function control word. You can choose both or any one. If choose both, over power or over current any one of them happen, the over load times will be accumulated.

Power limit can be set by user card. If the value is 0, cancel this function.

Current limit table: maximum 12 time interval and current limit each day. Support back up current limit table. If the value is 0, cancel this function.

Alarm function

Threshold alarm

When remaining credit or remaining day is less than the alarm threshold value, Low credit alarm LED will be flash for alarm. When the credit is run out, meter will trip out with continuously lighting low credit LED alarm. Then user can insert the user card for use the overdraft credit in configured, the overdraft value could be configurable by CPU card.

Meter has 3 levels alarm mechanism for low credit.

Lower than level1: flash frequency is 2s, Lower than level2: flash frequency is 1s, Lower than level3: flash frequency is 0.5s,

If without credit, the Low credit LED will be red,

Monthly consumption over limit alarm:

When monthly consumption over limit 1, LED flash, When monthly consumption over limit 2, LED flash, When remaining credit is use out, the user can continue use electricity when the meter open the following function:

1. Remain credit is use out when the meter is in friendly time, weekend, holiday, and the friendly overdraft threshold is more than 0 (can be set), the meter will not trip until the meter finish friendly time or the overdraft consumption reach to the friendly overdraft threshold. After trip, enter friendly time again, the meter will keep trip till new recharge.

2. Remain credit is use out when the meter is not in friendly time, weekend, holiday, but the overdraft threshold is more than 0 (can be set), the meter will trip and insert any card the relay close, and the meter will trip when the overdraft consumption reach to the overdraft threshold.

3. Monthly consumption over limit

If the remain credit >0 when the monthly consumption over limit2, deduct SF and other fees, remain credit ≤money active limit, meter permit overdraft consumption. When remain credit use out, even out friendly time, the meter permit overdraft consumption. Only when the overdraft reach to energy limit or time limit, the meter will trip.

Note: when the 3 items appear at the same time, only when 3 items satisfy trip condition.

Error alarm

If there is hardware failure when meter is on the running status, such as clock error, battery under voltage, cover open, etc. Relay will be open immediately till the error alarm is removed. Remove alarm is (remove the error and error code) only by the specified CPU card (which is the purchasing card and parameter setting card) which is authorized by vending department.

Priority: Hardware failure->Remaining credit->Remaining energy-> Overdraft credit-> Friendly credit -> Credit level

Error and CPU card error on LCD

(1) Error on LCD

No.	Display content	Error Type	Light	Relay	Solution to relieve error
01	oUEr-P	Over power limit	√	YES	Removed by card
02	oP-CouEr	Meter cover open	√	Optional	Removed by card
03	E-dATE	Clock error	√	Optional	Disappear when hardware error has removed
04	BATTEry	Battery under voltage	√	Optional	Removed by card
05	E-uOLT	Grid voltage is abnormal	√	Optional	Disappear when voltage error has removed
06	Cur-rE	Current reverse	√	NO	Removed by card
07	oP-TCou	Terminal cover open	√	YES	Removed by card
09	E-sTEAL	Tampering(Current unbalance)	√	NO	Removed by card
10	E-rELA4	Relay error	√	YES	Disappear when

					hardware error has removed
11	L-CrEdIT	Remaining credit is less	√	Optional	Disappear when the actual error has removed
14	non-CrEd	Remaining amount is less than or equal to 0	√	Optional	Disappear when the actual error has removed
15	P-odrAFT	Overdraft credit exceeds the credit threshold	√	YES	Disappear when the actual error has removed
16	P-FrIEnd	Overdraft credit exceeds the friendly threshold	√	YES	Disappear when the actual error has removed
17	L-LEu1	Remaining credit is less than the first alarm threshold	√	Optional	Disappear when the actual error has removed
18	L-LEu2	Remaining credit is less than the second alarm threshold	√	Optional	Disappear when the actual error has removed
19	L-LEu3	Remaining credit is less than the third alarm threshold	√	Optional	Disappear when the actual error has removed
20	E-LoADd	Data loading failed	No	Optional	Disappear when the actual error has removed
22	OUEr XX	Monthly consumption over limit 1	√	No	After billing error disappear automatically
23	dEbtor	Monthly consumption over limit 2 and overdraft	√	YES	Disappear after recharge

Note: All the errors can be removed only when the actual errors have been removed, even the error can be removed by cards.

(2) CPU Card error

Error Code	Error
00	Card type is wrong
01	No response information from the card, maybe the card or the card operation interface is broken
02	Response information can be transmitted from the card, but not the card type for the meter
03	Physical type of card is right, but the meter does not support the logic operation of this card
01	The corresponding information of meter and card is wrong
01	User number error
02	Vending system serial number error
03	The card cannot be accepted for the current logic state of meter
04	Meter is not compatible with the data structure version of card
05	What inside the card is electricity quantity, but the meter is time meter, which rejects electricity and accepts energy charge only
06	The card has been occupied, please open a new open account card
07	The meter has been occupied
08	Meter number error
02	Data is abnormal (Data format or check information error)
01	Data format error
02	Check information error
03	No operation authorization
04	Write-back area is full
05	Setting area over length
0300	Password verification is failed
0301	Out of expiry date

0400	Vending times error
0500	Card is pulled out in advance, operation is not completed
0600	The seal button is not pressed when operating the card
0700	Overrun error, the recharge credit plus the remaining credit is greater than the preset "hoarding credit threshold"
0701	Data loading error, cannot accept any card
08	Error related to CPU card
00	Card reset error
01	File select error
02	Authentication inside card error
03	Authentication outside card error
04	Write data error
05	Read data error
06	File length definition error

Note: LCD display "-good-" when insert the right CPU card
 LCD display "-USED-" when repeatedly insert "open account card", "vending card" or "trouble shooting card".

7 Card functions

7.1.1 Open account card and Credit recharge card (Customer card)

Open account for user, and take user information and system information to the meter at the first time.

It is used for credit recharging and hold by user.

It also supports to configurable the related prepayment parameters.

Only the meters that in user mode can accept this card.

When open account can be configured by software check meter number, replace meter number or not change the meter number.

7.1.2 Mending card

Vending station can mend a new card for users if they loss or broke the user card.

1. When recharge times in the card is 1 more than the recharge times in the meter, the meter accept the card normally.
2. When recharge times in the card equal to the recharge times in the meter
 - 1) The money in the card is less than the last recharge money which meter record, meter will accept the card and display 'USED'
 - 2) The money in the card is more than the last recharge money which meter record, meter will accept the card. And recharge time will not change, recharge money = money in card - the last recharge money which meter record.

7.1.3 Account cancellation card

Account cancellation card is used for closing an account and returning the credit. The card is changed user card, so as to read the current remaining credit of meter and clear it. Only the meter that under User state can accept the card and meter enters into installation status after successful processing.

7.1.4 Reset card

Without modifying the data in the meter, the card can be accepted in both all state. After successful processing, meter status is changed to the manufacturing status.

7.1.5 Parameter setting card

The card is used for setting related parameters and can be accepted in any logic state.

If using parameters card to remove alarm, it will has card number. This number will be record into meter when remove alarm. Then this number also is feedback to S/W by user card. The parameter card has expire date limit, if the time is over expire date, meter can't accept this card.

This card can be configured to check meter number or not check meter number.

7.1.6 Detection and reading card

The card supports reading the relevant prepayment parameters, data and records in the meter. One card can correspond to multiple meters; data items that need reading are configurable, and the meter amount that can be read is based on the reading data amount.

7.1.7 Big capacity reading card

The card supports reading the relevant prepayment parameters, data and records in the meter. One card can correspond to multiple meters, data items that need reading are configurable, and the meter amount that can be read is based on the reading data amount (maximum 250 meters).

7.1.8 Setting card in factory

Setting whole parameters which about the prepaid function to meter.

This card can use in any meters with the communal password, without check of system number and user number. It can return data from meters. Operation of event record is reset event record.

7.1.9 Clear meter card of factory

It is used to clear data and all events records.

This card can use in any meters with the communal password, without check of system number and user number. It cannot return data from meters.

7.1.10 Setting card when out of factory

It can set the parameters for the prepaid function of meters. The data including remain energy set or clear, to clear all of data of the meters that as history data or events records.

It can change the meter state from the manufacturing and inspection state to installation state.

7.1.11 Relay control card

This card can be used in any mode. Meter will change the relay status after insert this card. The meter number will be return to card when success operating.

The card has expire date limit, if the time is over expire date, meter can't accept this card.

The card can be invalid after the programmable time.

After inset this card, LCD will add ' r-ICC Ard' 'invalid time' in auto cycle display.

7.1.12 initialize parameter card

Factory mode and installation mode this card can be used without limitation, user mode the card can be used only one time.

7.1.14 read and tariff setting card

Using this card can read data about prepayment data, parameter and record from meter. Tariff setting: step, tariff, tariff offset, special tariff.

7.1.15 big capacity change meter card

This card can be used many times. This card without comparing system ID and user ID. When inserting card need press program key, only when meters have the same EEPROM code, can copy data. When change meter use the new meter number open account, at the same time save the old meter number in new meter, so the meter can accept both new user card and old user card.

7.1.16 user card read and tariff setting card

This card will check system number and has expire date.

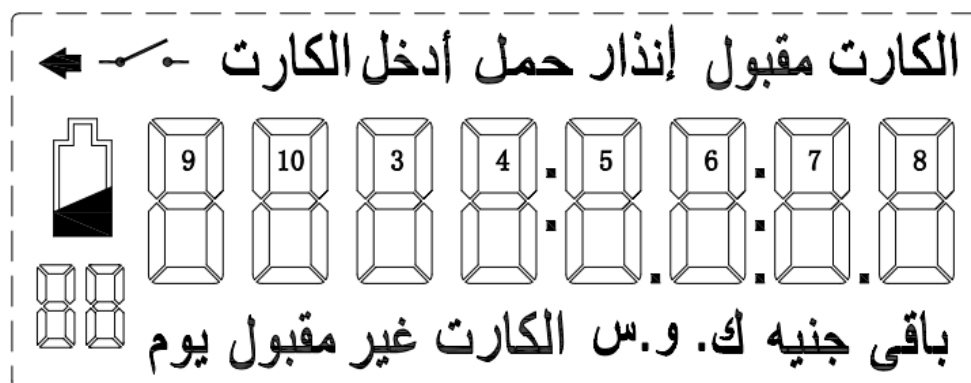
Tariff setting: This card can save some tariff scheme, including step, tariff, tariff offset, and tariff offset 2. The card will copy the same tariff scheme number content to the meter.

7.1.13 Installation card

This card like the parameter setting card, but only can be used one time in user mode.

1 Meter display

LCD Full display



LCD display content

After power on, LCD screen shows full screen, then displays version number. Full screen display keeps for 2 seconds by default, press the cyclic button to enter into normal display, skipping full screen display. There are 2 kinds of LCD display mode: auto-cycling display, button query display.

When meter is powered on, if there is no button press, no card insert action in 5 minutes, the LCD screen will close. Besides, the LCD is closed under low power consumption condition, press the button or insert the card can enable the LCD display.

1.1.1 Auto-cycling Display

After running for a while (default time is 3 seconds, configurable), meter will automatically switch to next screen display. The cycling display items are configurable, and 16 items can be configured at most.

1.1.2 Button query cycling Display

Press the query button, meter will display next item. Display sequence, display screen number, and display items (60 items can be configured at most) are all configurable.

The default set is shown as below :

Note: The time display format is 12 hours, distinguished by “A” (morning) and “P” (afternoon), which displays as the first letter in the left of the screen.

Sequence number	Display Item Name
01	Remaining energy (0.01kWh)
02	Remaining credit Negative: display minus mark, 3 decimals and 4 integers Forward: 5 integers and 3 decimals.
03	Total accumulative energy consumption(kWh)(accumulate since opening account)
04	Energy consumption of current month (0.01kWh) (monthly consumption)
05	Total energy consumption of last month (0.01kWh) (monthly consumption)
06	Current credit consumption (0.001 local currency)(It will display energy consumption of the current month if in energy metering mode)

07	Current tariff No.
08	Current limit(A)
09	Remaining energy alarm threshold(0.01kWh)
0A	Tariff (00 : step tariff 01 : TOU tariff 02 : energy)
0B	Over voltage threshold (%UA)
0C	Under voltage threshold (%UA)
0D	Current demand
0E	Billing day (date : time)
0F	Remaining credit alarm threshold (0.001 local currency)
1X	step value (X = 0 ~ 8) (0.01kWh)
1F	Current tariff value
2X	Tariff value (X = 0 ~ 9) (0.001 local currency/kWh)
2A	Loading date of new tariff
2B	Active date of new tariff
2C	Starting date of current tariff
2D	Compensation value of current tariff
2E	Compensation value of next step tariff
2F	Current tariff schedule number
30	Current date
31	Current time
32	Week, Sunday ~ Saturday (0,1,2,3,4,5,6)
33	Purchased credit of current month (0.001 local currency)
34	Purchased credit of last month (0.001 local currency)
35	Current power (“L “means fire line metering and “N “means neutral line metering)
36	Power demand
37	Overdraft threshold
38	Friendly threshold
39	Power off times

3A	Running time after power on (Minutes)
3B	Running time after reset (Minutes)
3C	Energy consumption after power on (kWh)
3D	Remaining days (days) (not appear
3E	Remaining days threshold
3F	Accumulative purchased credit(0.001 local currency)
40	Meter cover open times
41	Total energy consumption of last 1 month 6 integers , 2 decimals XXXXXX.XX 0.01kWh
42	Total energy consumption of last 2 months 6 integers , 2 decimals XXXXXX.XX 0.01kWh
...	...
52	Total energy consumption of last 12 month 6 integers , 2 decimals XXXXXX.XX 0.01kWh
53	Reverse energy
54	Unbalanced energy
55	Starting date of latest tamper event
56	Starting time of latest tamper event
57	Ending date of last tamper event
58	Ending time of last tamper event
59	Terminal cover open times
5A	The total overdraft energy
5B	The total overdraft money
5C	MD(A) of current month
5D	MD(kWh) of current month
5E	Consumption credit of relay fault
5F	Fee of zero consumption monthly
60	Energy consumption because of relay error
61	The first level alarm threshold (0.001 local currency/kWh)
62	The second level alarm threshold (0.001 local currency/kWh)

63	The third level alarm threshold (0.001 local currency/kWh)
64	Hoarding threshold (local currency)
65	Friendly time start
66	Friendly time end
67	Energy consumption of last 1 day
68	Energy consumption of Last 1 hour
6B	Weekend day
6C	Weekend day
6D	Weekend day
6E	Weekend day
6F	Holiday If today is holiday, the LCD display the date of today; If today is not holiday, the LCD display the date of the next holiday
70	Pulse constant
71	00 : Factory mode 01:installation mode 02: user mode
72	System type number (xxxx)
73	8 low bits of user number (xxxxxx)
74	8 high bits of user number (xxxxxx)
75	8 low bits of meter number (xxxxxx)
76	8 high bits of meter number (xxxxxx)
77	IC CArd (IC - CPU Card meter)
78	Measuring firmware version
79	Relay control status(00: control by meter logic, 01:remote control by system 02:Control by disconnect CPU card)
7A	The remain times of relay automatically close when over power limit
7B	Current month MD(power) happen time(DD:HH:MM)
7C	Current month MD(A)happen time(DD:HH:MM)
8x	(x=1-4) starting time and meter number of 1-4 time zone format: Month and Day-time frame

9x	(x=1-8) starting time and tariff number of current time frame , format :Hour and minute - tariff number
Ax	(x=1-4)1-4 tariff value
Bx	(x=1-4)1-4 energy consumption of the tariff
C1	Voltage
C2	Current (Loop A)
C3	Current (Loop B)
C4	Power (Loop A)
C5	Power (Loop B))
C6	Power factor
C7	IEC61107
C8	Closing times of Relay
C9	Trip-out times of Relay
CA	Accepted recharge card times
CB	Rejected recharge card times
DX	(x=0-B) 0-B, starting time of power limit time phase. Format is Hour Minute
EX	(x=0-B) 0-B, power limit of current time phase. Format is Power
F1	Current reverse occurred date of last time Year-Month-Day
F2	Current reverse occurred time of last time Hour Minute Second
F3	Meter cover open occurred date of last time Year-Month-Day
F4	Meter cover open occurred time of last time Hour Minute Second
F5	Terminal cover open occurred date of last time Year-Month-Day
F6	Terminal cover open occurred time of last time Hour Minute Second
F7	Date of the latest credit recharge Year-Month-Day
F8	Time of the latest credit recharge Hour Minute Second
F9	Date of the latest relay error Year-Month-Day

FA	Time of the latest relay error Hour Minute Second
FF	State of terminal cover, and top cover 1means open and 0 means closed

1.1.3 Overdraft credit icon display

When total remaining money is 0, user can insert the function card to overdraft credit when there is overdraft limit. The LCD screen displays minus icon “-”, and the rest 6 characters are displayed as the overdraft money (3 decimals). (For example, when the total overdraft money is 6, the screen will display -6.000)

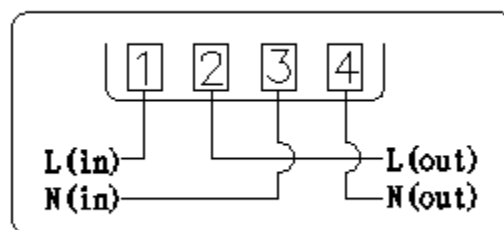
2 Outline and installation requirement

After qualified inspection and being packed with seals before leaving factory, meter can be installed for utilization.

Meter should be installed in distribution box or block terminal, with the height ranges from 0.8 to 1.8 meters, and without corrosive gas in the air.

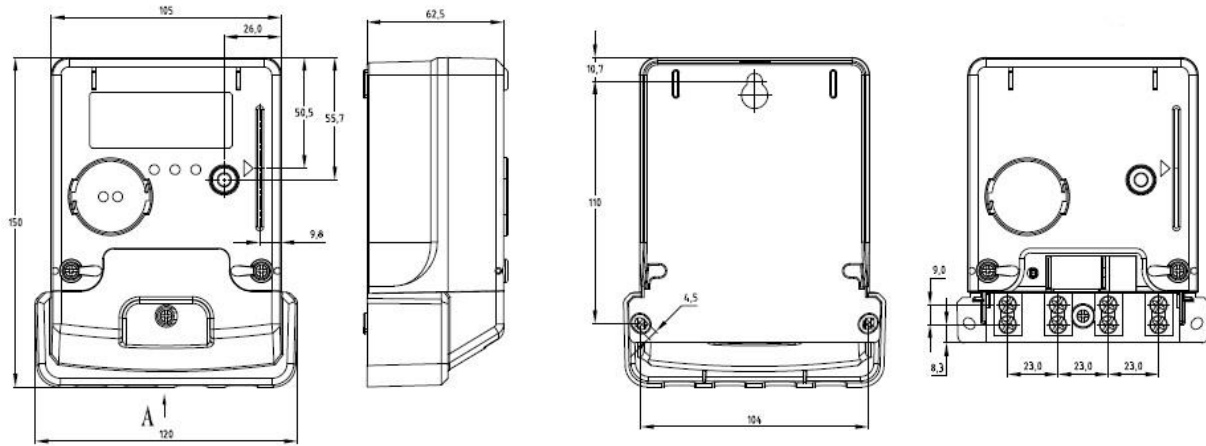
Alternating connection of meter should conform to the basic craft requirements:

- 2.1.1 If the conductor is too thick, several wires should be reduced from the conductor so as to connect to meter terminal box.
- 2.1.2 If the conductor is far smaller than the terminal hole, more wires should be added to the conductor before connecting.
- 2.1.3 Wire connection



【Figure 9.1】 Wiring connection diagram

2.1.4 Outline and installation dimension



【Figure 9.2】

3 Transportation and Storage

There should be no severe impact during meter transportation and stripping, according to the general technical specification stipulation of meter packaging.

Meter should be saved in the original package, and the environmental temperature around the preserved place is $-25^{\circ}\text{C} \sim +55^{\circ}\text{C}$, meanwhile, the relative annual average humidity does not exceed 85%, and without corrosive gas in the air.

Meter should be put on bench when saved in the warehouse, and the stacked cases number does not exceed 10. After stripping, the maximum stacked number of packaged meters is 10.